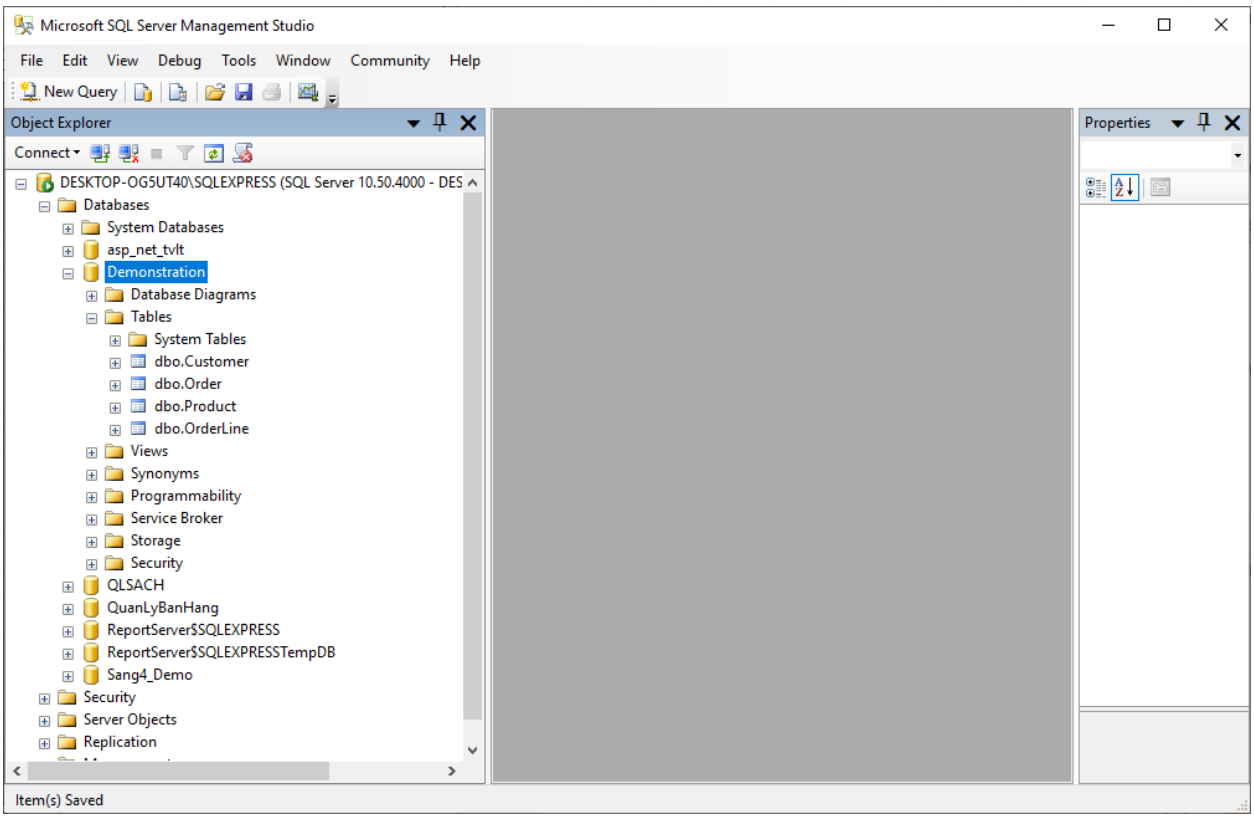
Create a database in SQL server

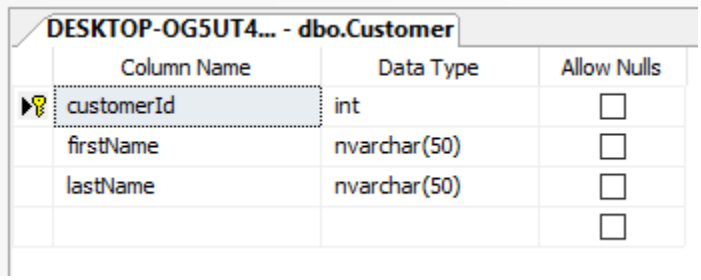
1. Create a new database with 4 tables

Database name: Demonstration

4 tables:

- Customer
- Order
- OrderLine
- Product





DESKTOP-OG5UT4... - dbo.Customer

Column Name	Data Type	Allow Nulls
customerId	int	☐
firstName	nvarchar(50)	☐
lastName	nvarchar(50)	☐
		☐

DESKTOP-OG5UT40...ion - dbo.Order

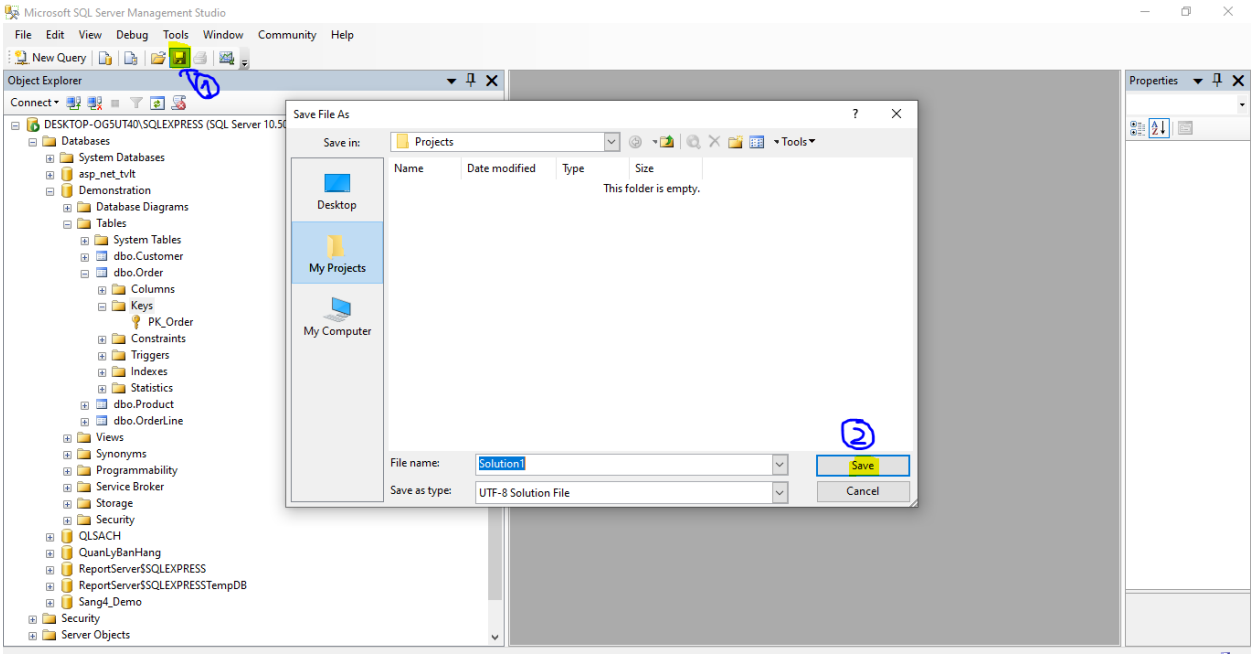
Column Name	Data Type	Allow Nulls
orderId	int	☐
orderDate	datetime	☑
customerId	int	☑
		☐

DESKTOP-OG5UT40...n - dbo.Product*

Column Name	Data Type	Allow Nulls
productId	int	☐
description	nvarchar(50)	☐
		☐

DESKTOP-OG5UT40...- dbo.OrderLine*

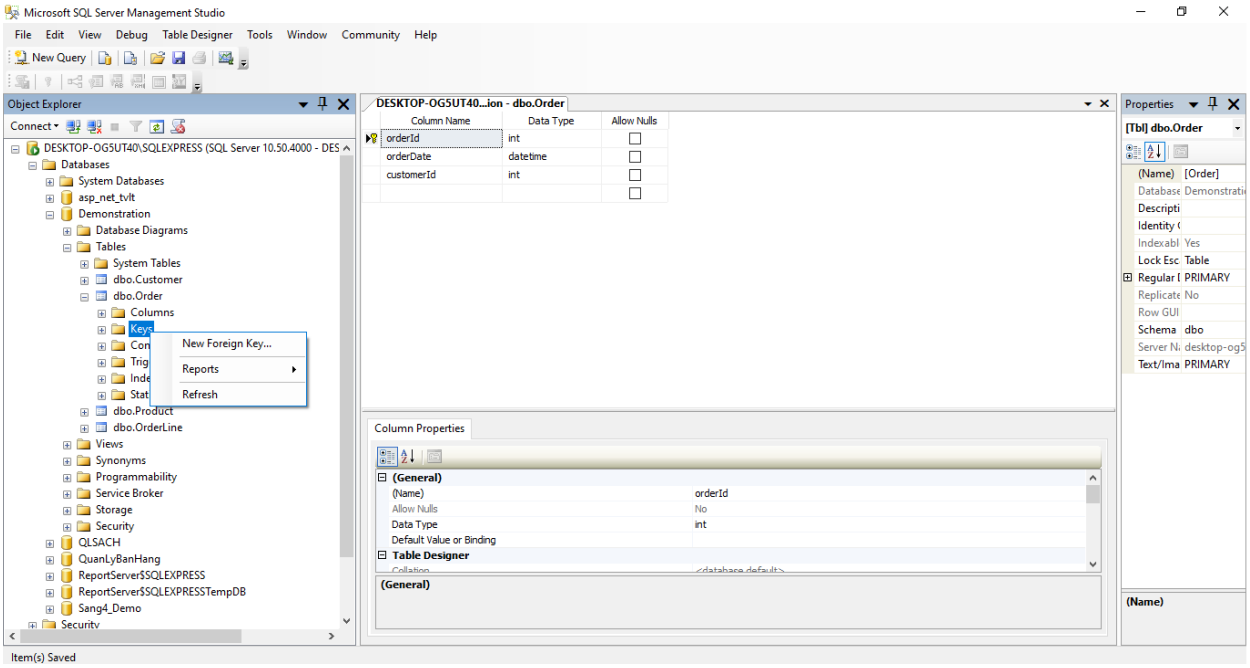
Column Name	Data Type	Allow Nulls
orderId	int	☐
productId	int	☐
quantity	int	☐
		☐

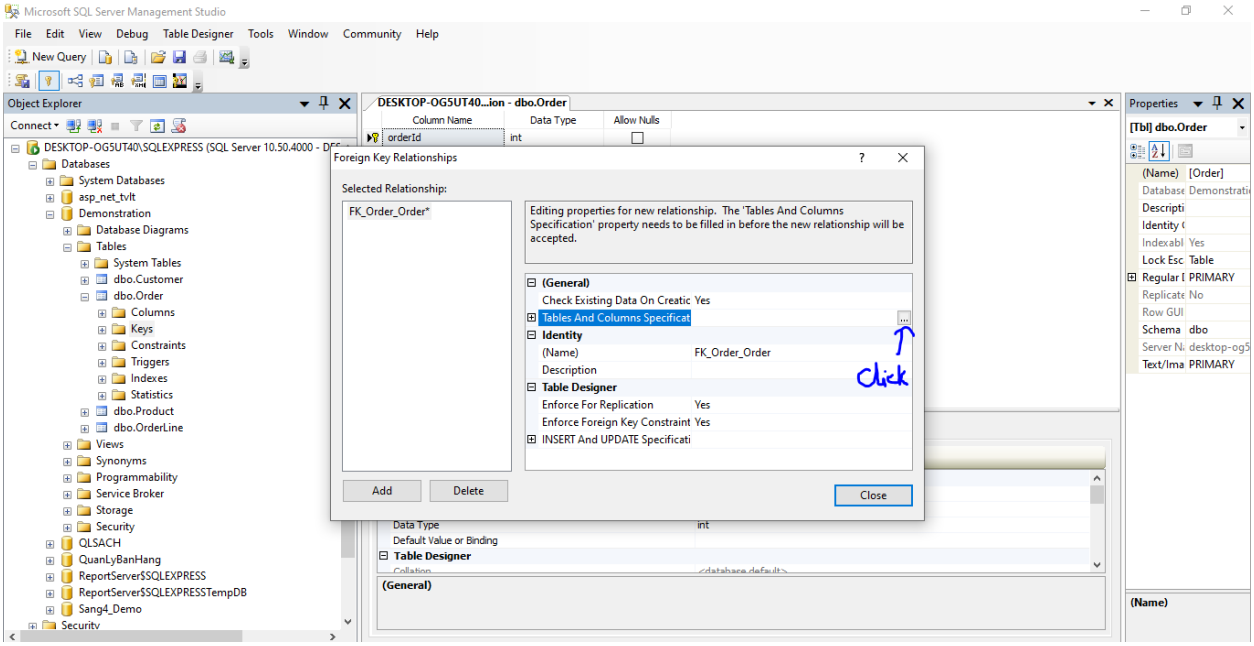


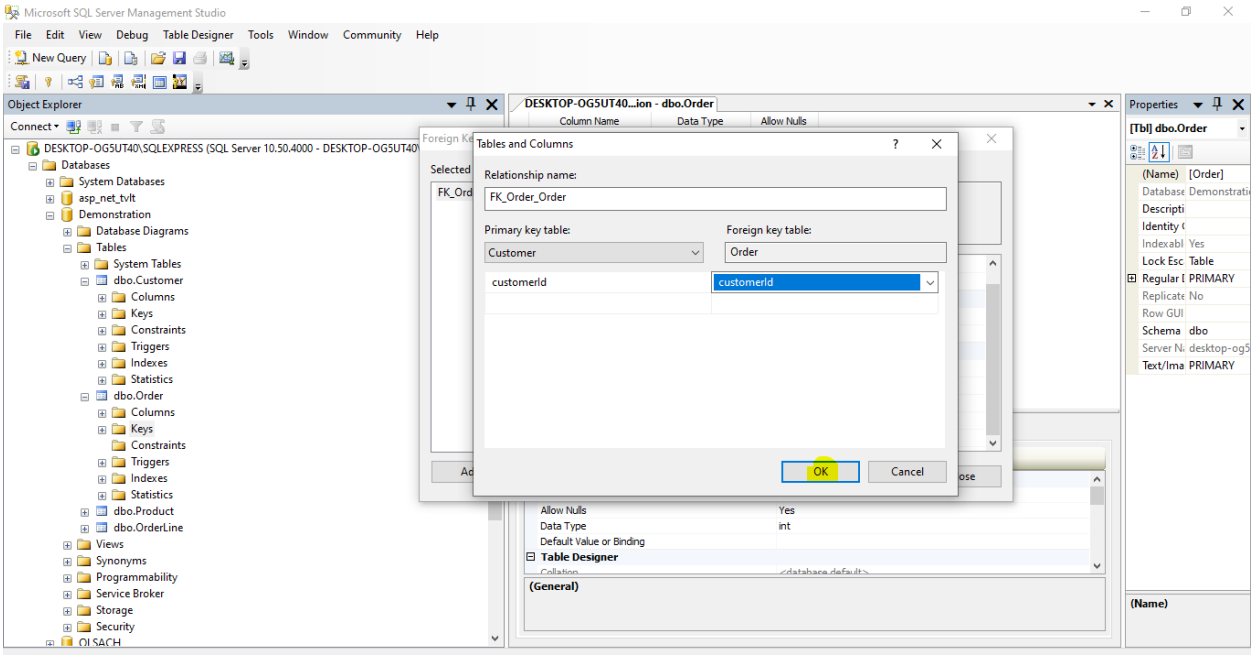
2. Create a relationship between two tables

2.1. The first way of using the designer

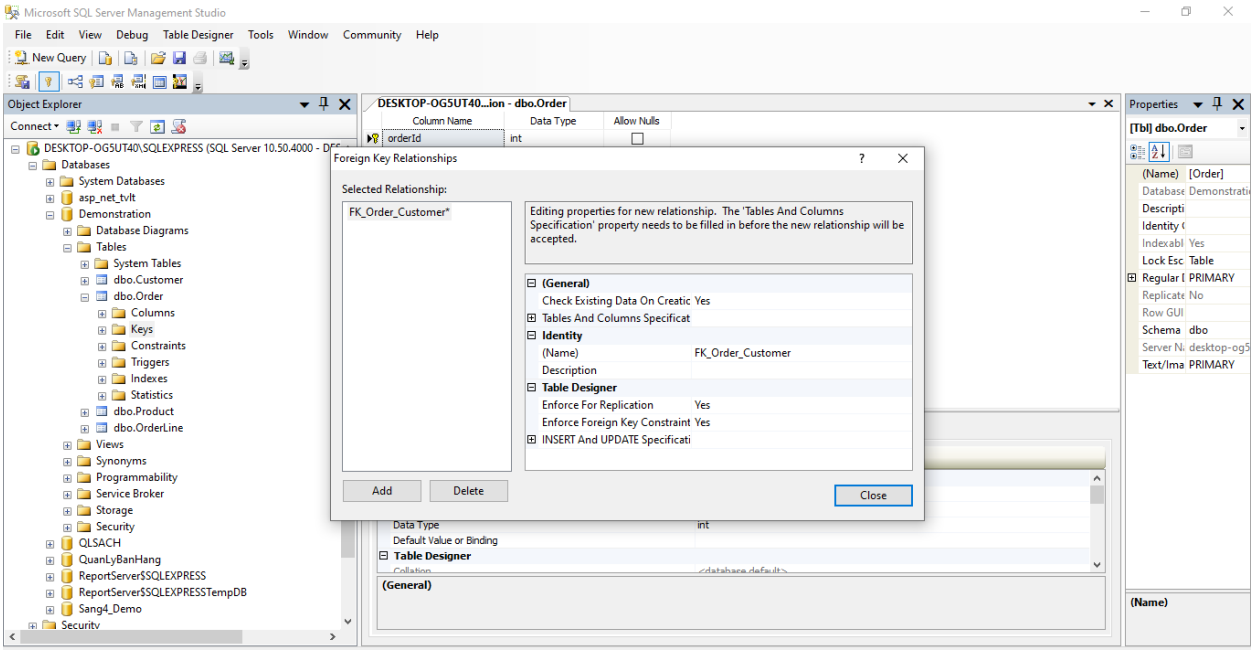
1-n between Customer and Order tables: a customer can play many orders.

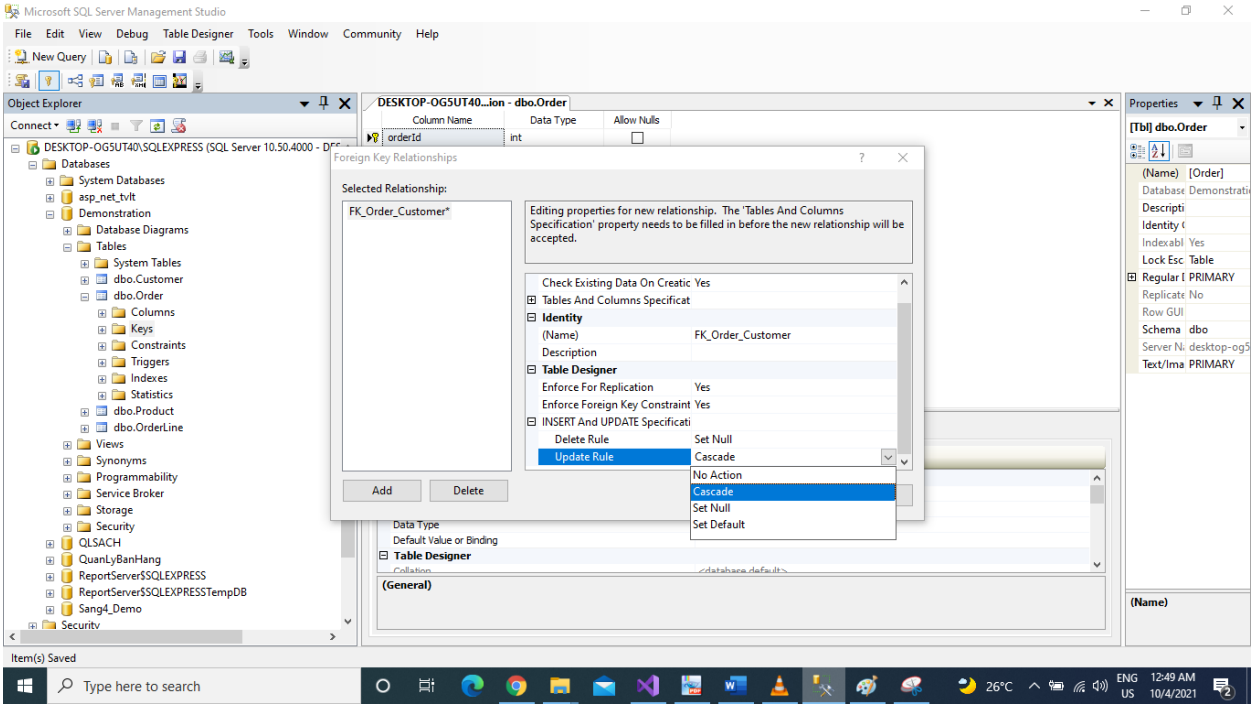


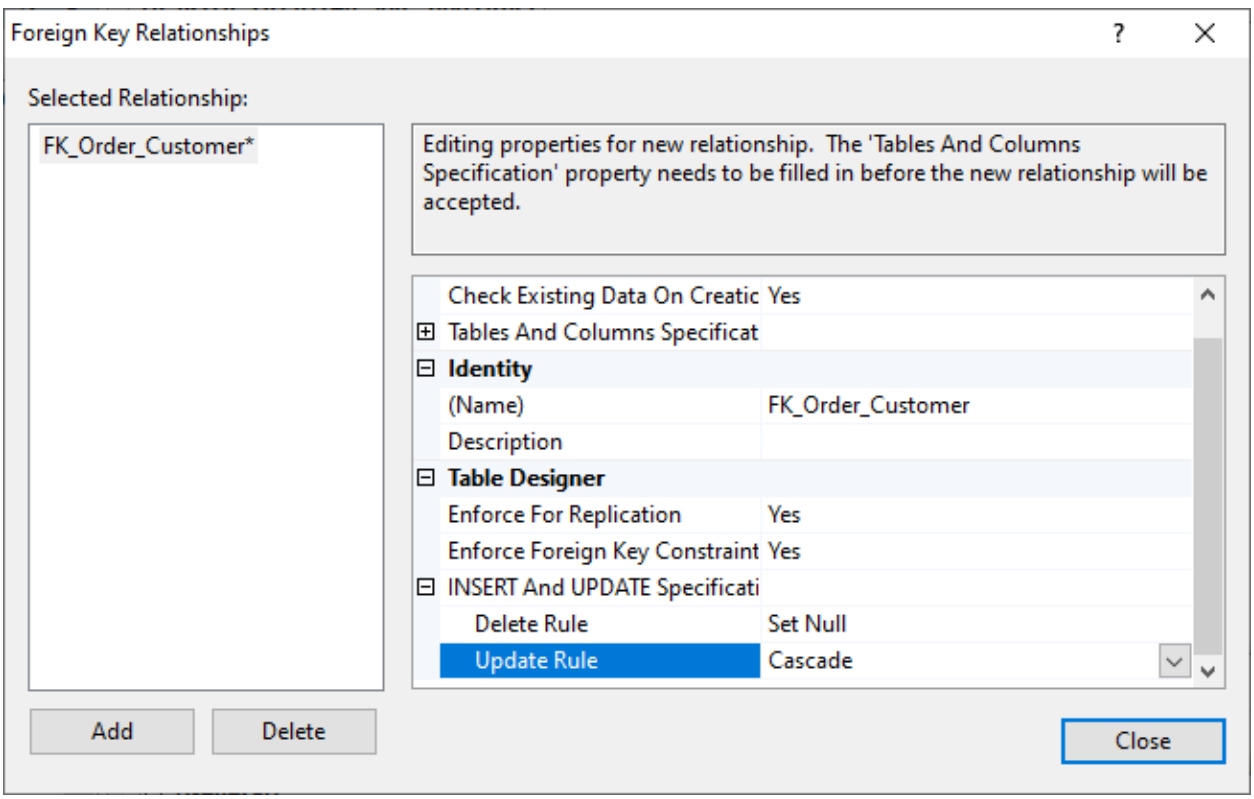




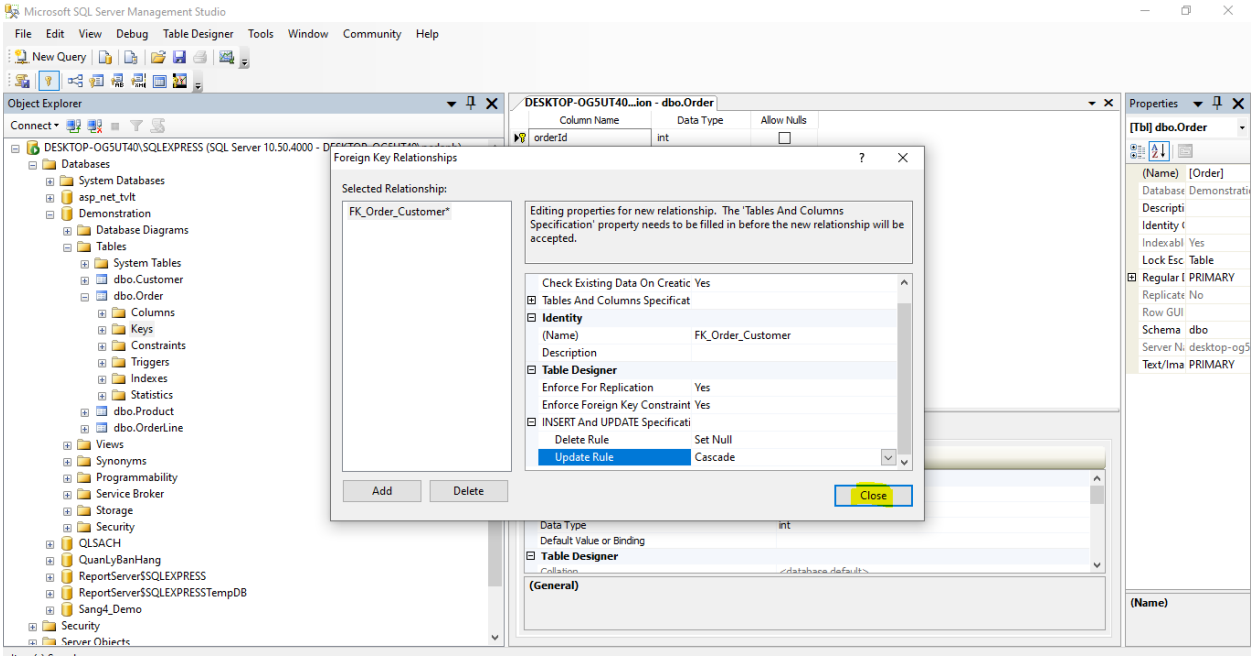
(Relationship name: FK_Order_Customer)







(We should set: Delete Rule: Set Null and Update Rule: Cascade)



(Close)

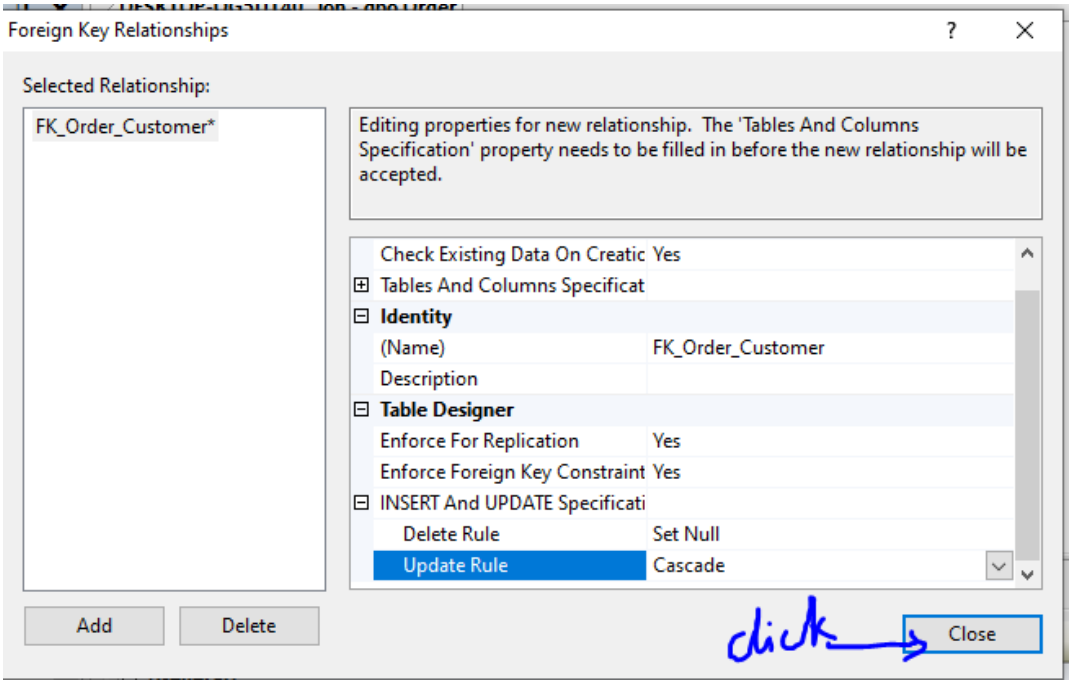
Delete rule:

If deleting happens on the parent table, e.g., Customer, what happens on the child table, e.g., Order table? Because we have the foreign key relationship between two tables Order and Customer, if we delete a customer on the Customer table, some records on the Order table doesn't point to an actual record on the Customer table; or we delete a customer on the Customer table and then add a new record for that customer but use a new customerId, we will get wrong associated customerIds on the Order table. That is realy not good, so we need to do something to handle that.

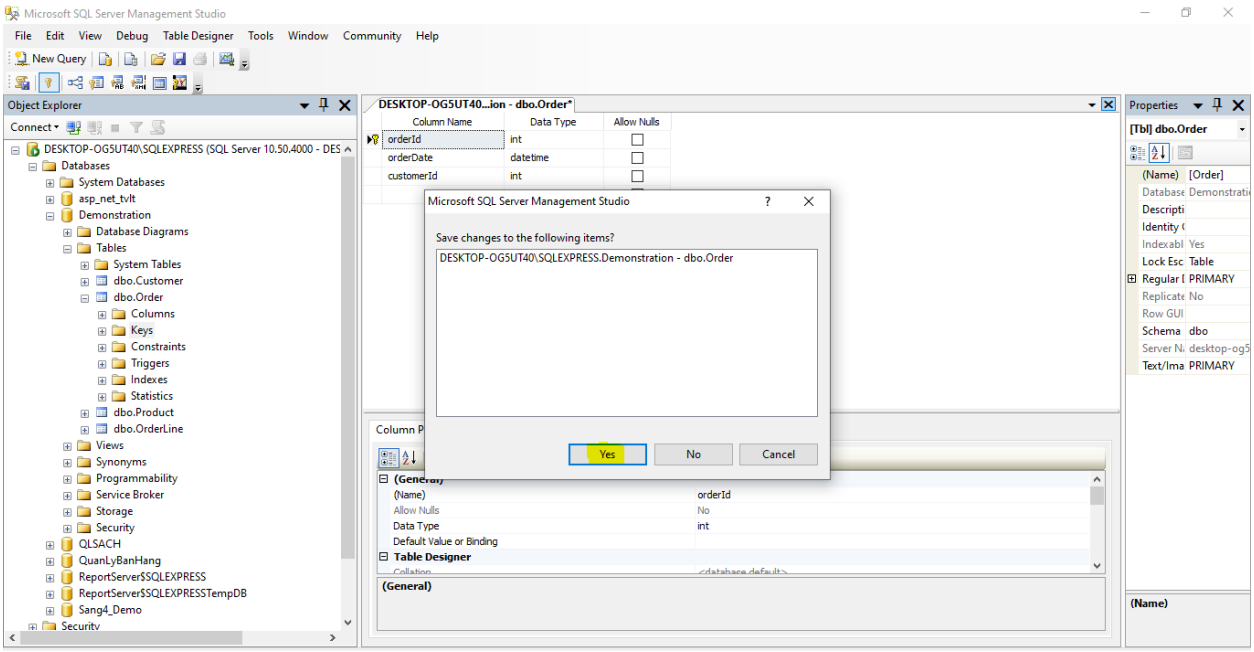
Delete rule shows what happens to the Order table (all records on the Order table) where the foreign key values point to the record on the Customer table.

We have some options for deleting a customer on the Customer table:

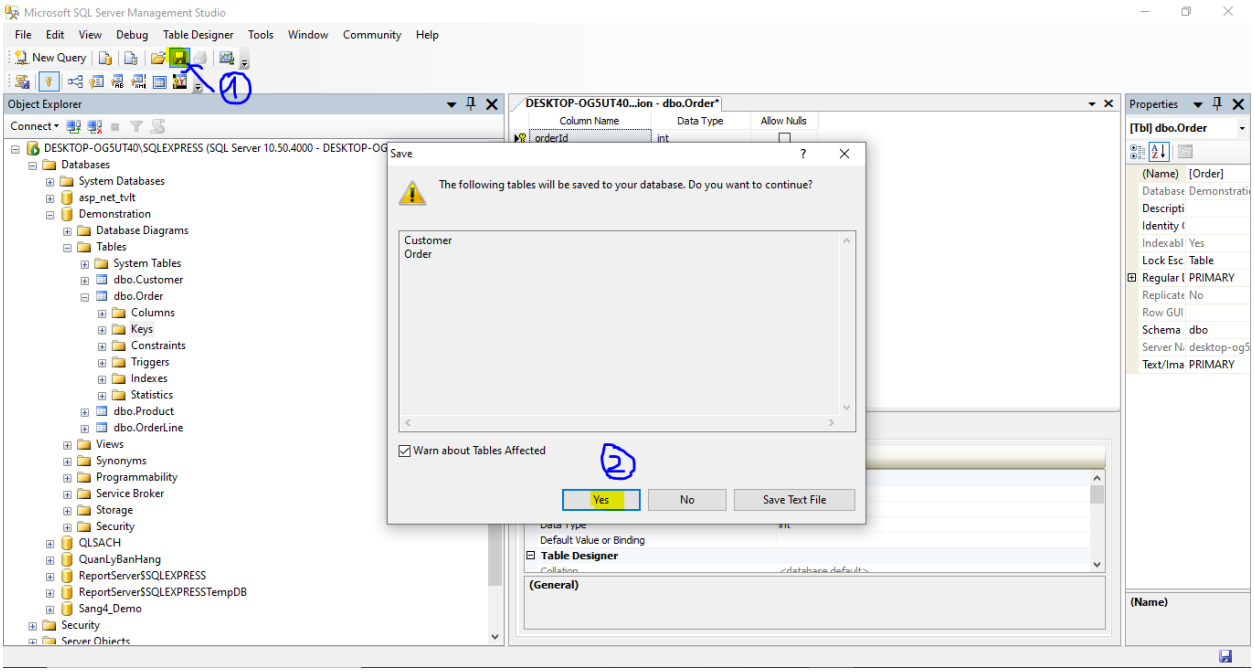
- No action: do nothing.
- Cascade: delete all the orders related to the that customer on the Order table.
- Set Null: still keep the orders (on the Order table) but set the values of the customerId of related records on the Order table to Null. This is probably the best solution here.
- Set Default: still keep the orders (on the Order table) but set the values of the customerId of related records on the Order table to a default value, e.g., 0, 100, etc.



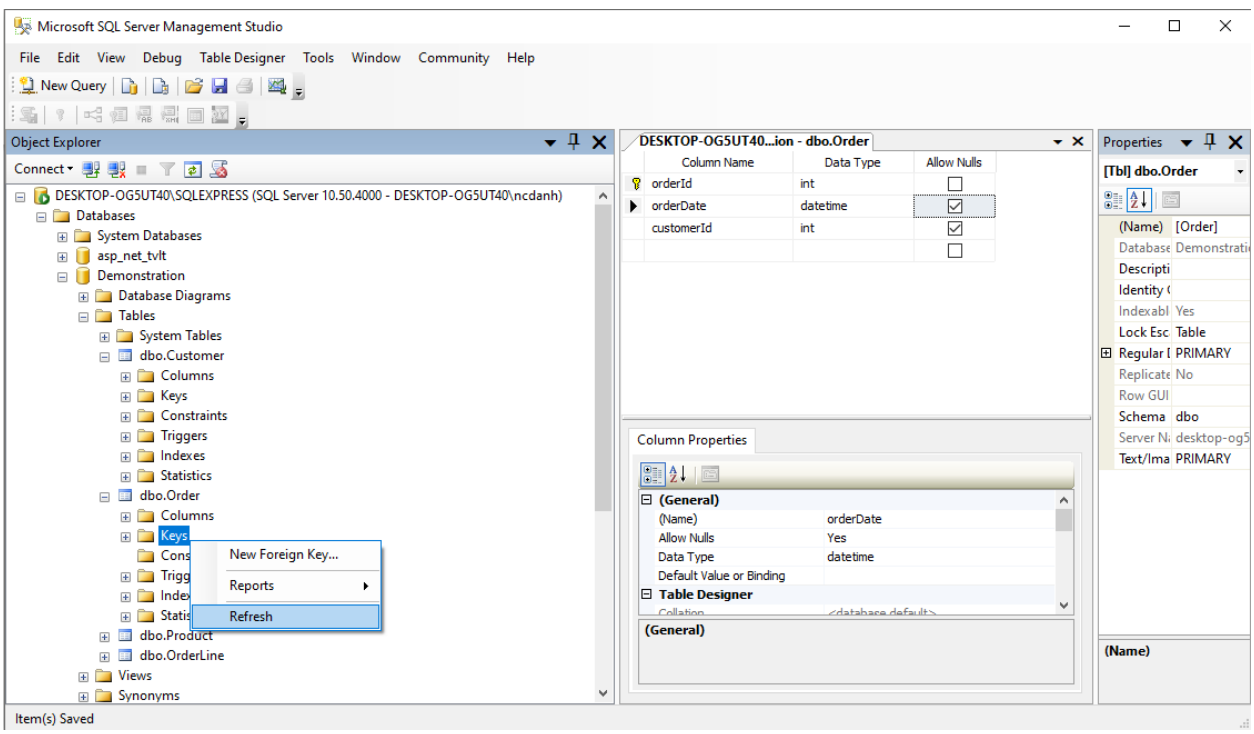
(Close)



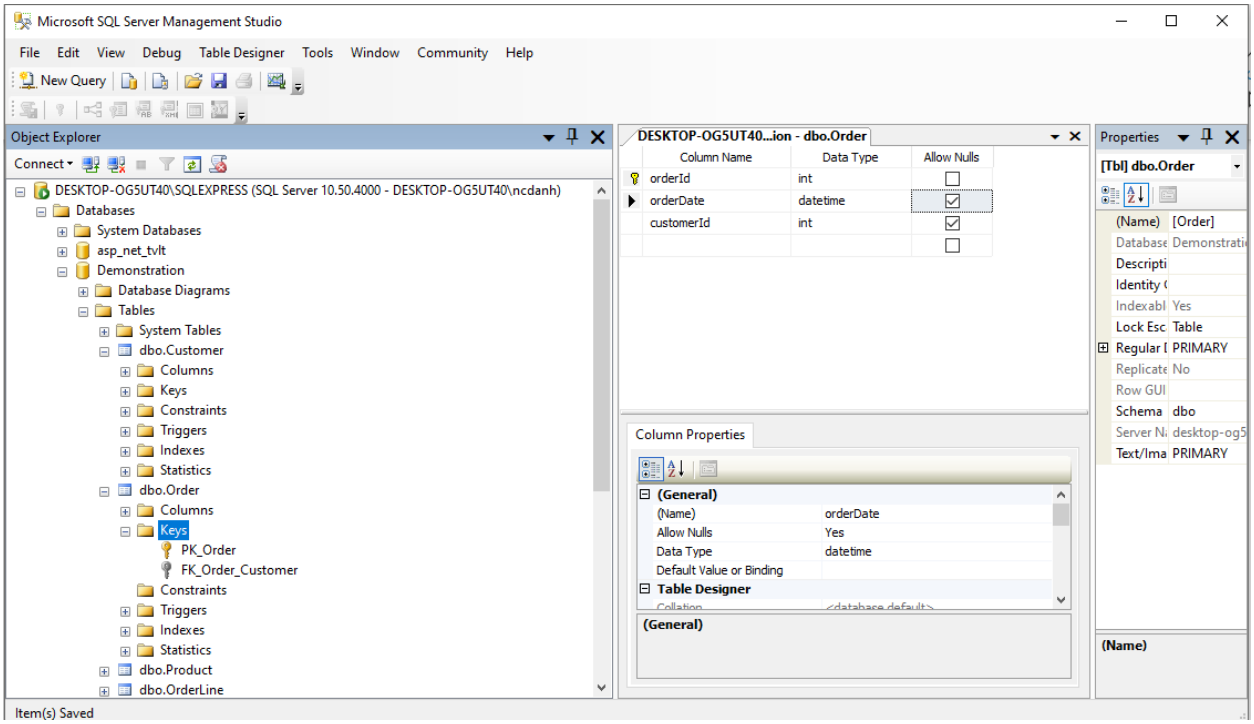
(Save the Order table => Yes)



(SQL Server saves two tables because both the tables have been changed)



(Refresh)

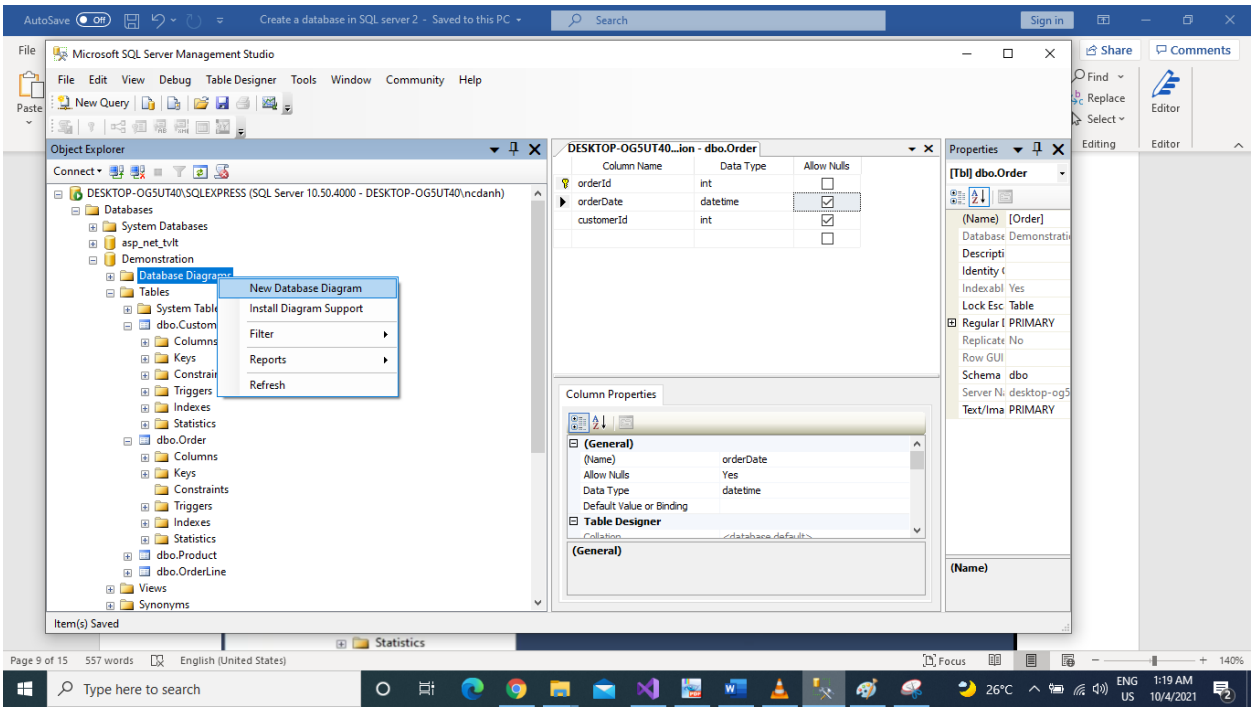


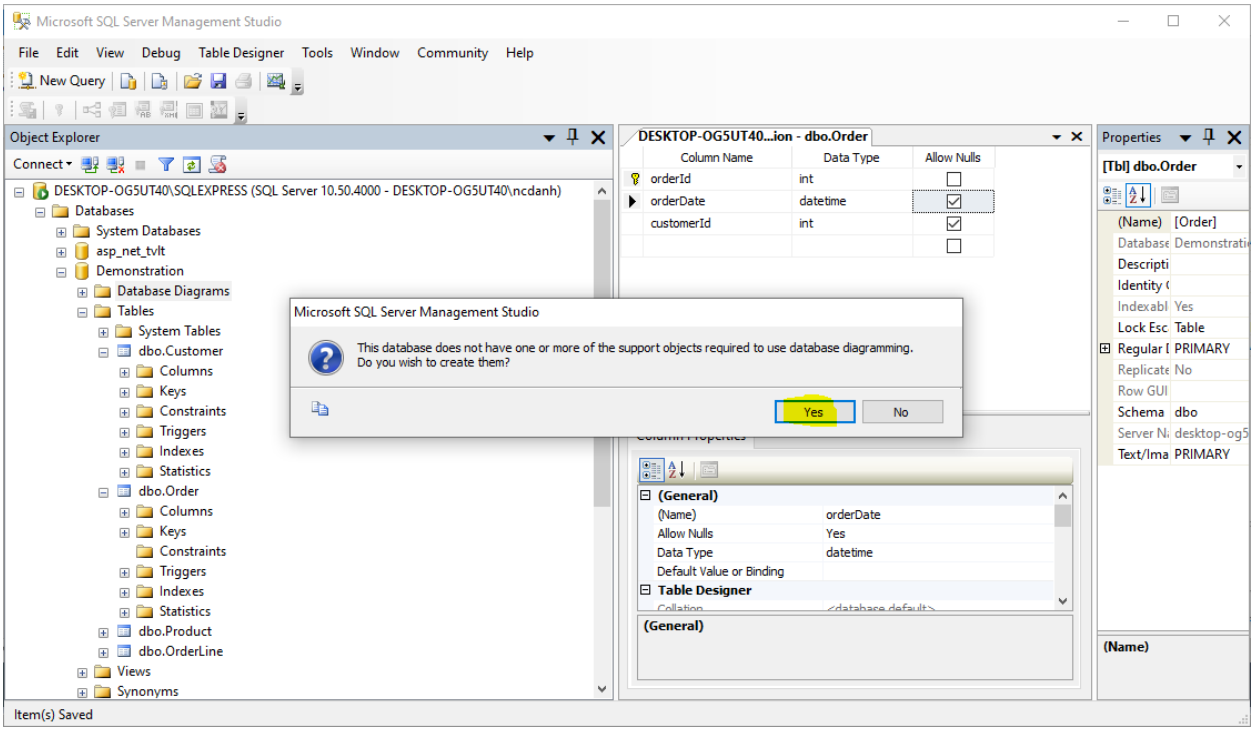
(SQL Server has not updated the new items on the left => Refresh)

After refreshing, we can see the Order table has two keys: Primary key and Foreign key that links the Order table and the Customer table together.

2.2. The second way of using the designer

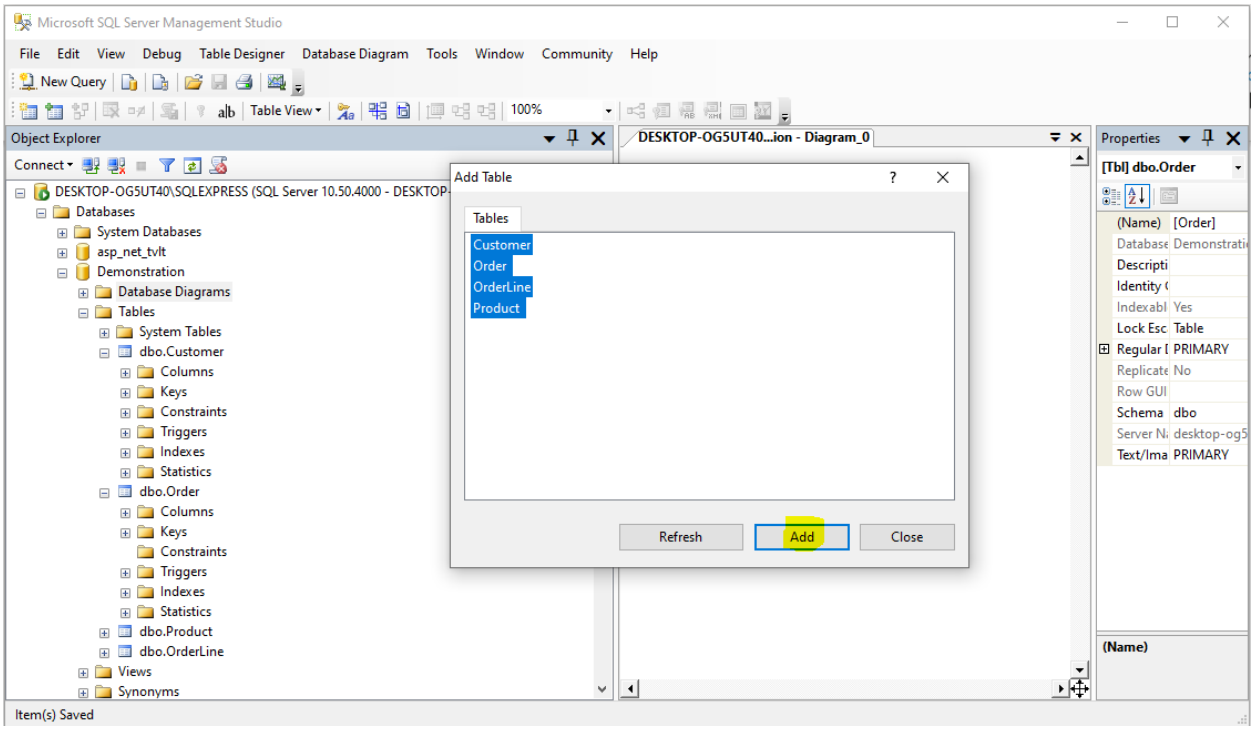
Using Database Diagrams



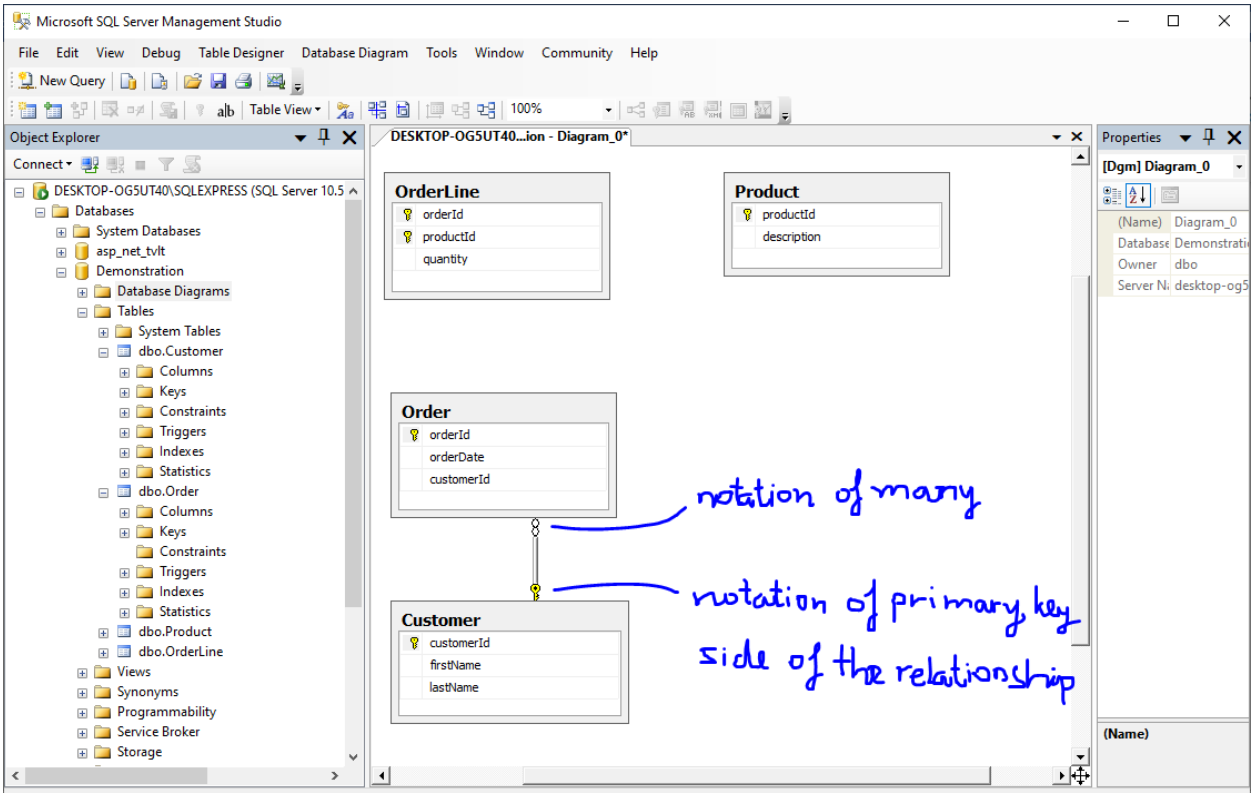


(Yes)

(New Database Diagram)



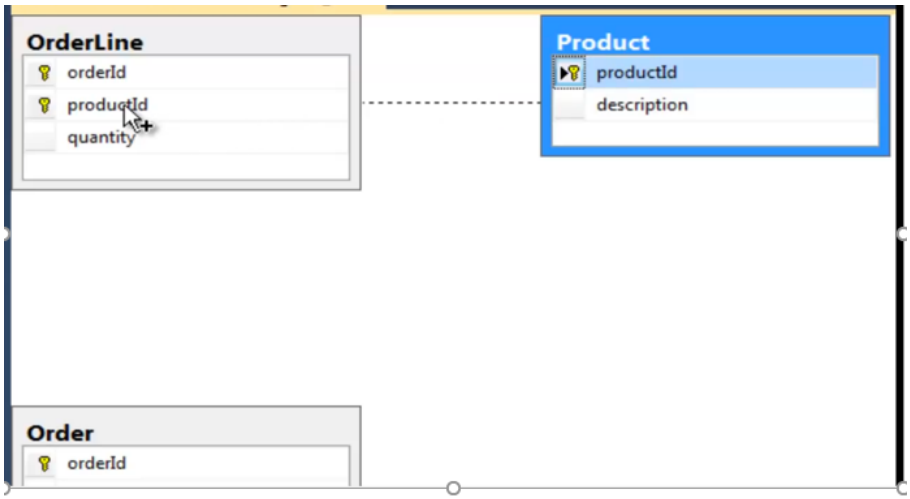
(Select all thetables => Add)

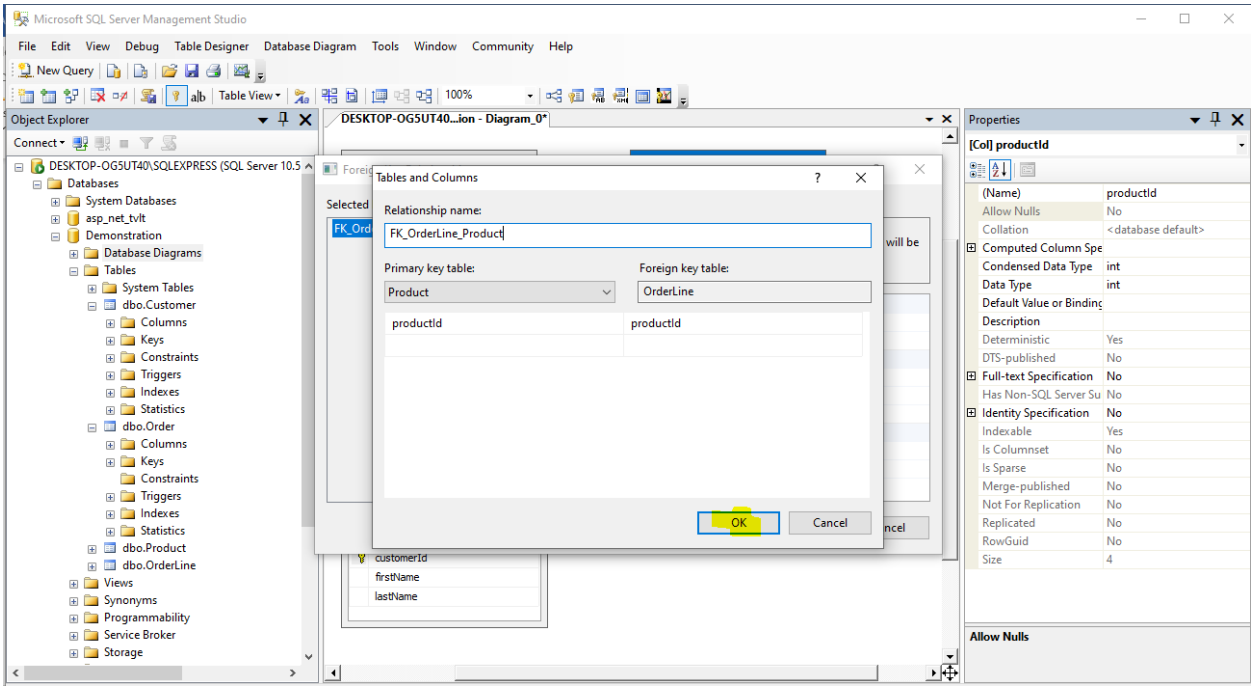


(There exists the relationship between the Customer and Order tables)

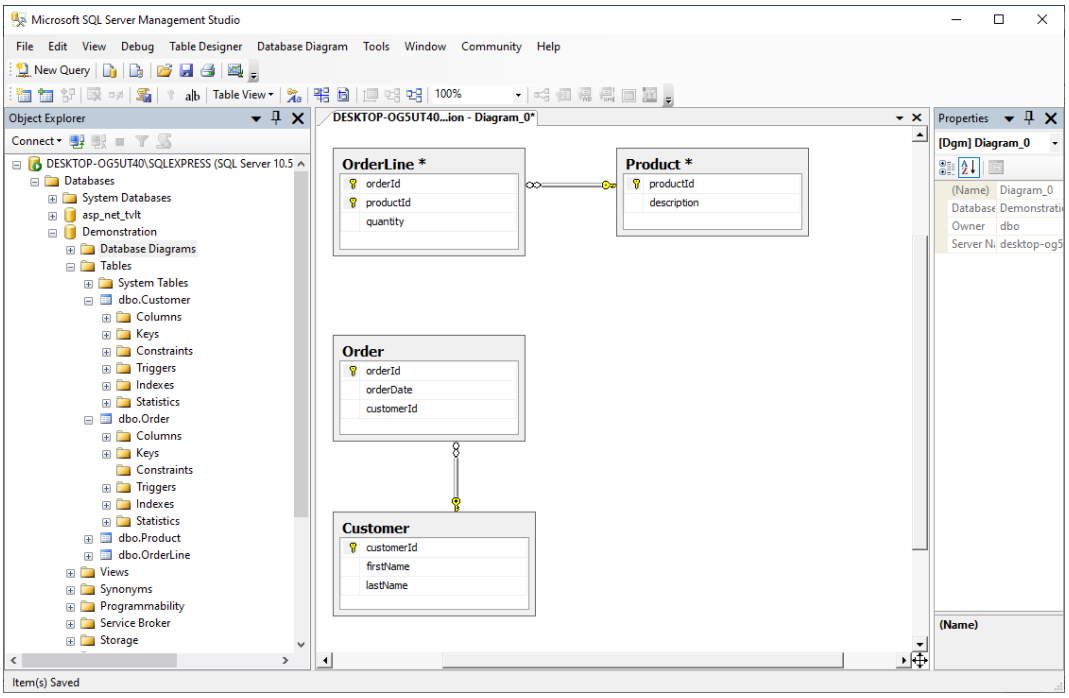
Now we make the relationship between the Product and Orderline tables. How to do this?

Select and hold down the left button of the mouse on the primary key productId of the Product table and drag to the top of the productId attribute of the Orderline table and release the mouse.

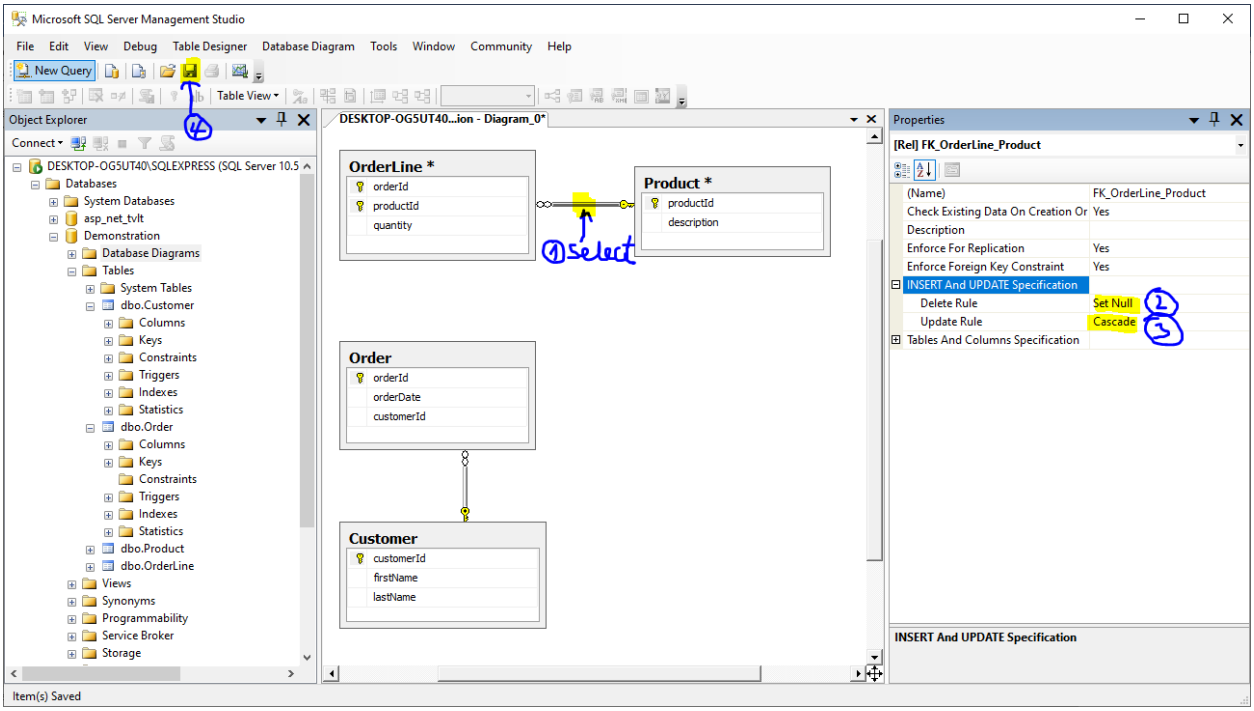




(OK => OK)



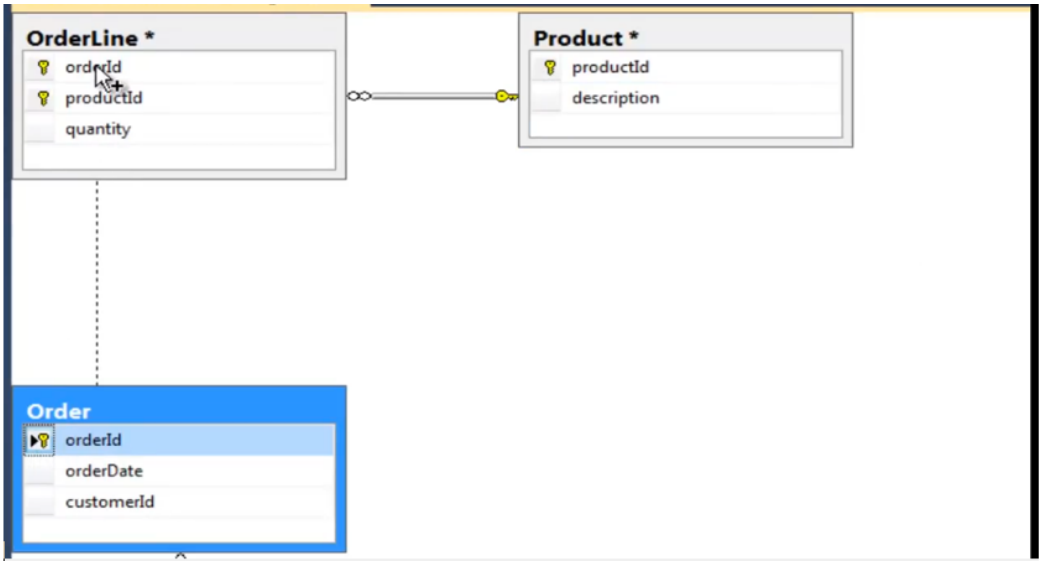
(A relationship is created)

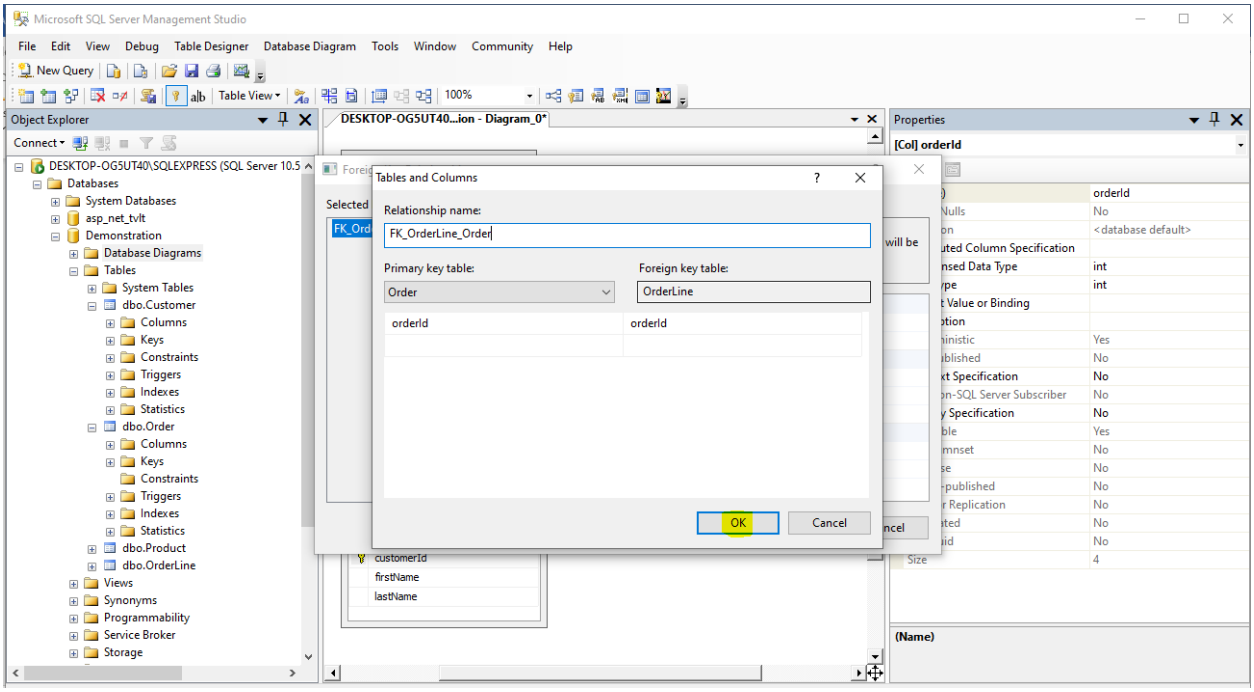


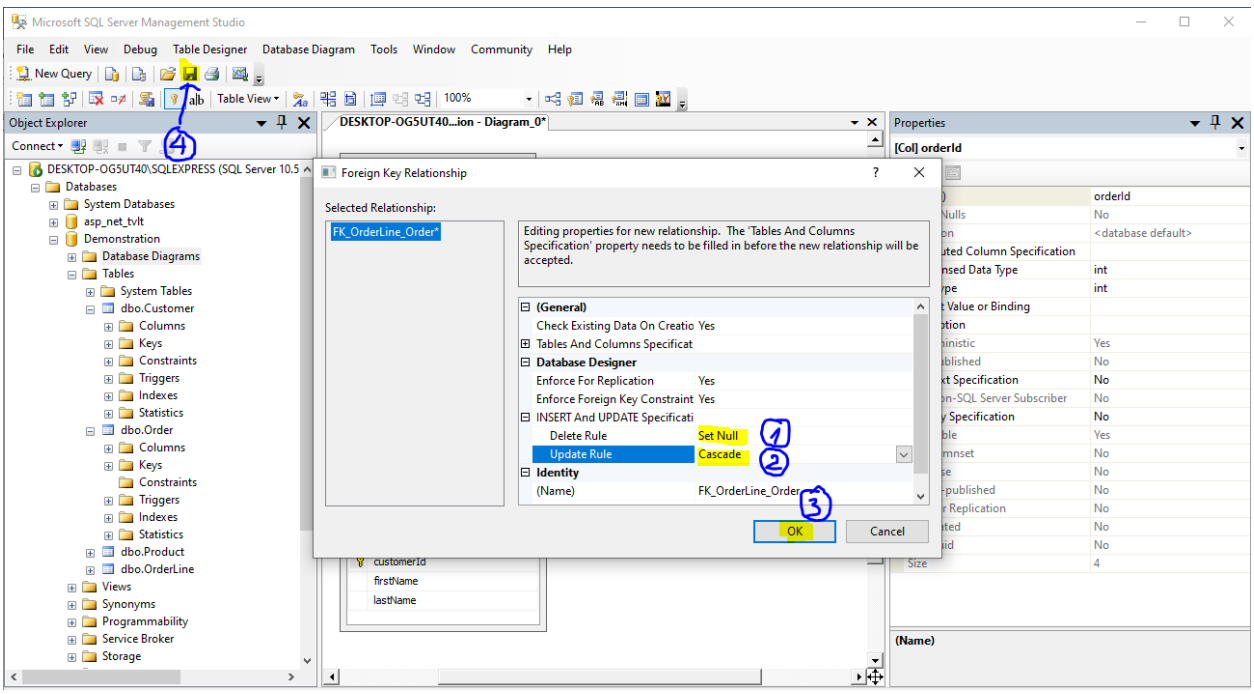
(Set Delete Rule and Update Rule)

Now we make the relationship between the Order and Orderline tables. How to do this?

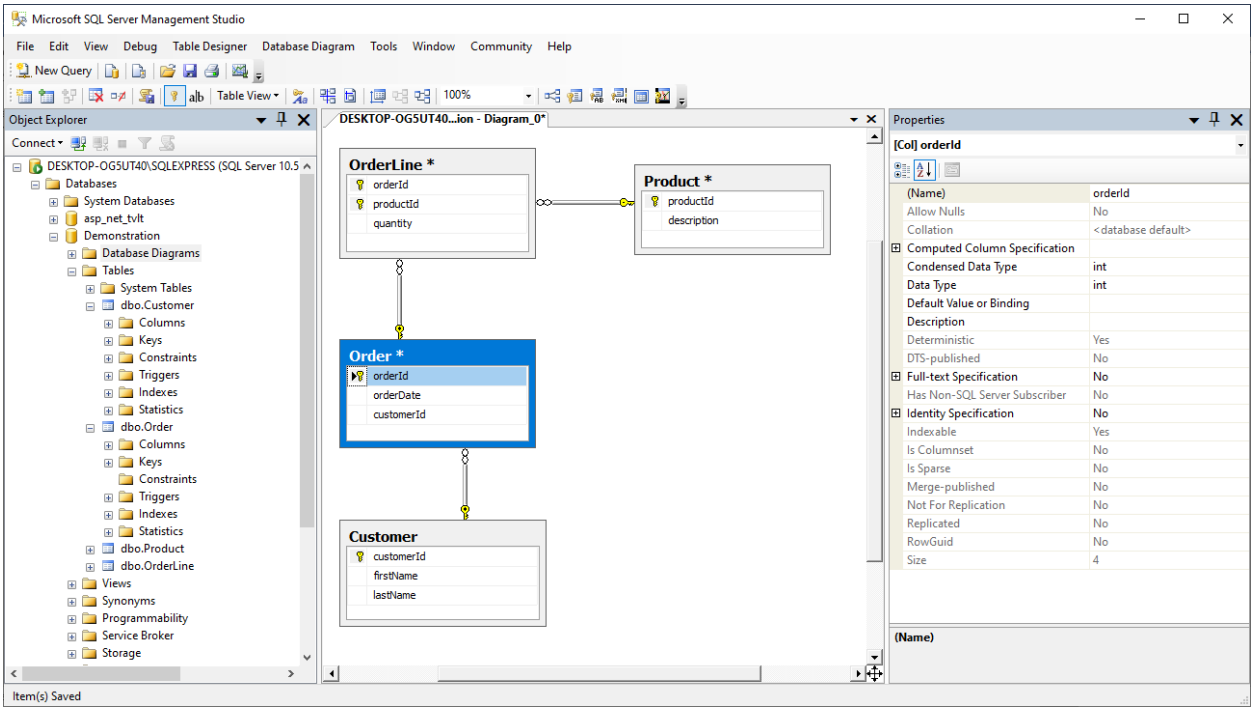
Select and hold down the left button of the mouse on the primary key orderId of the Order table and drag to the top of the orderId attribute of the Orderline table and release the mouse.



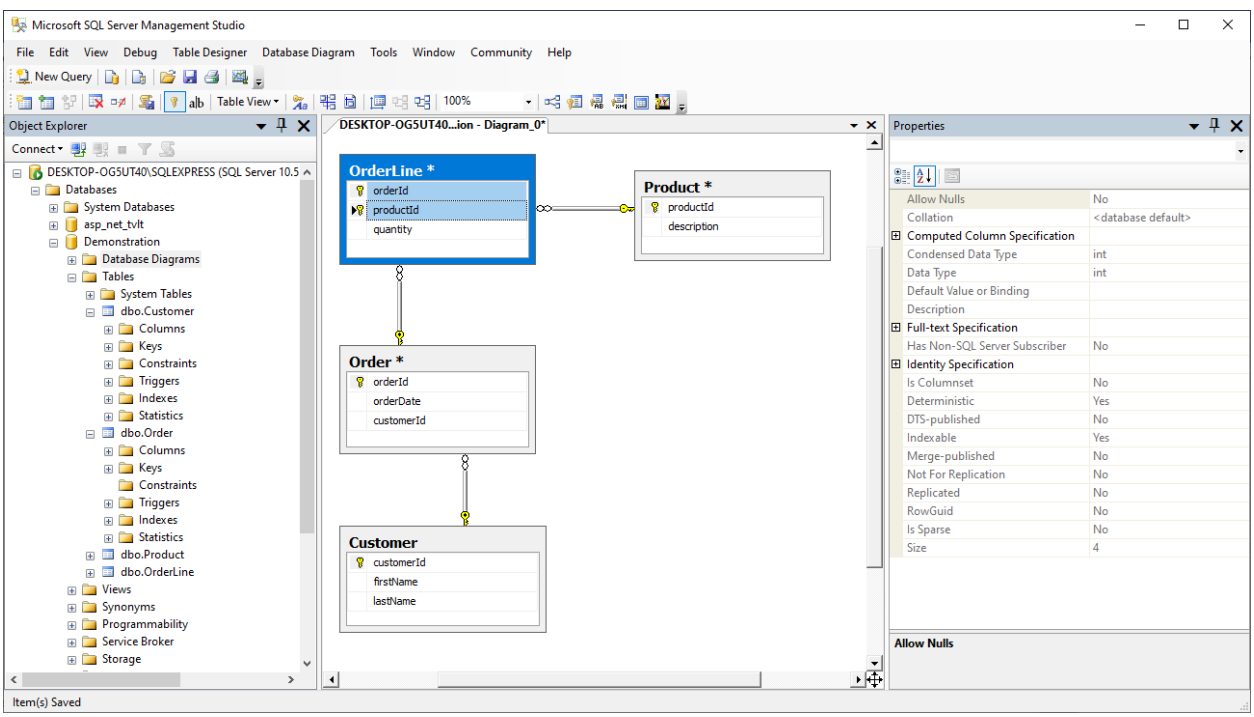




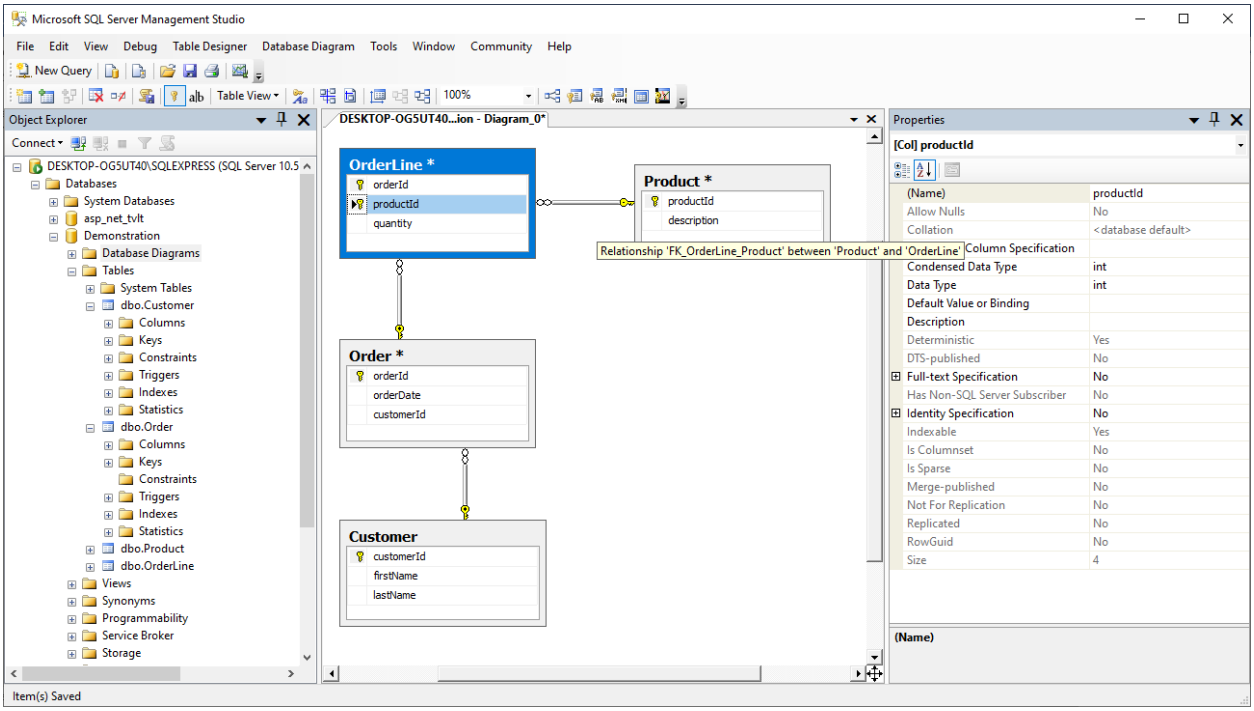
(OK => OK => The relationship is established)



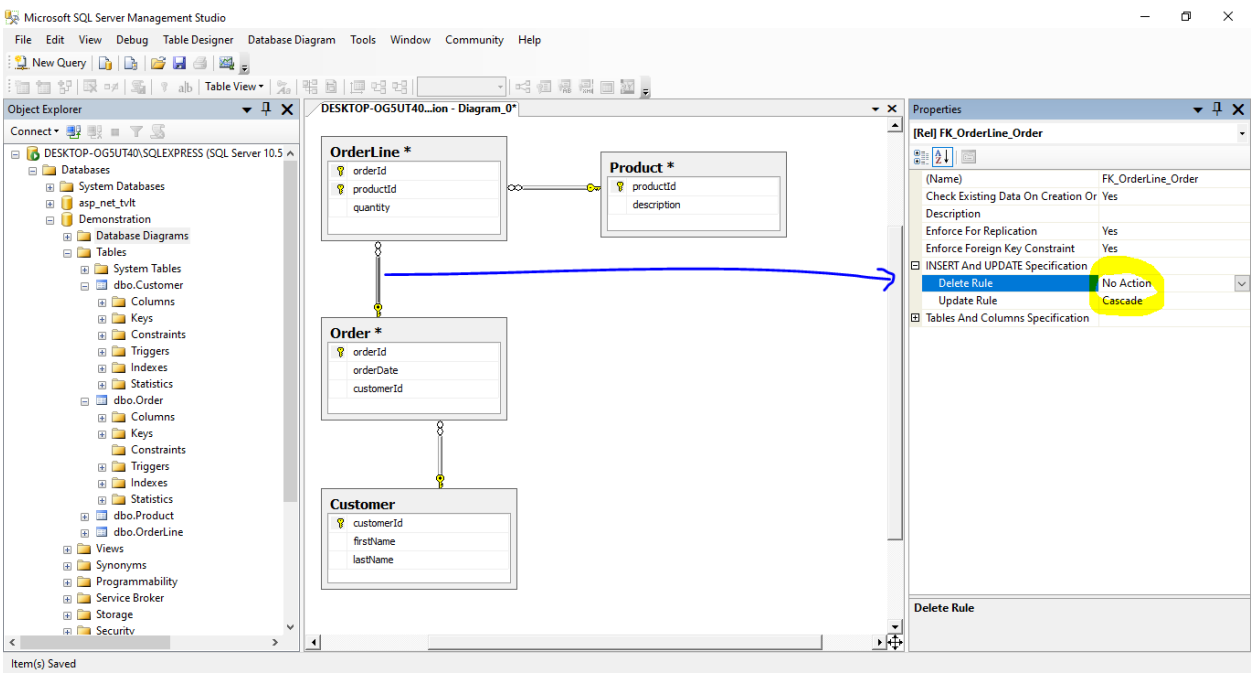
It's important to note that both the orderId and productId compose to create the primary key of the OrderLine table. However, individually each attibute searches a foreign key link back to parent table: the productId on the OrderLine table to the productId on the Product table; the orderId on the OrderLine table connects to the orderId on the Order table.

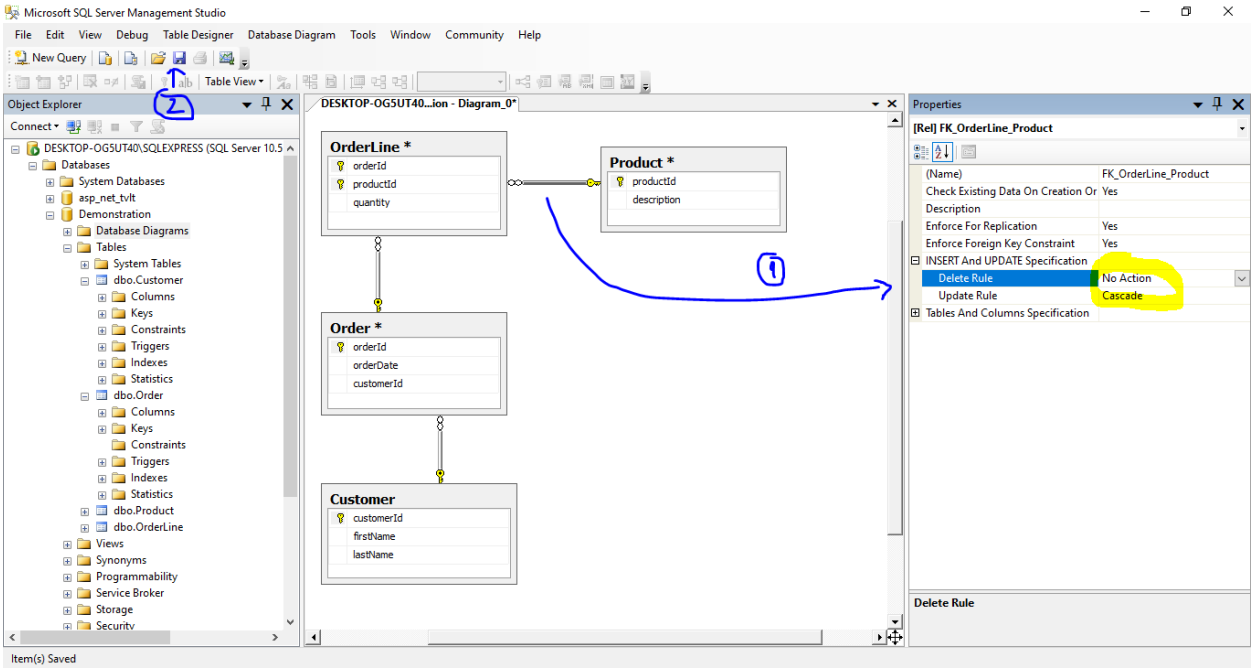


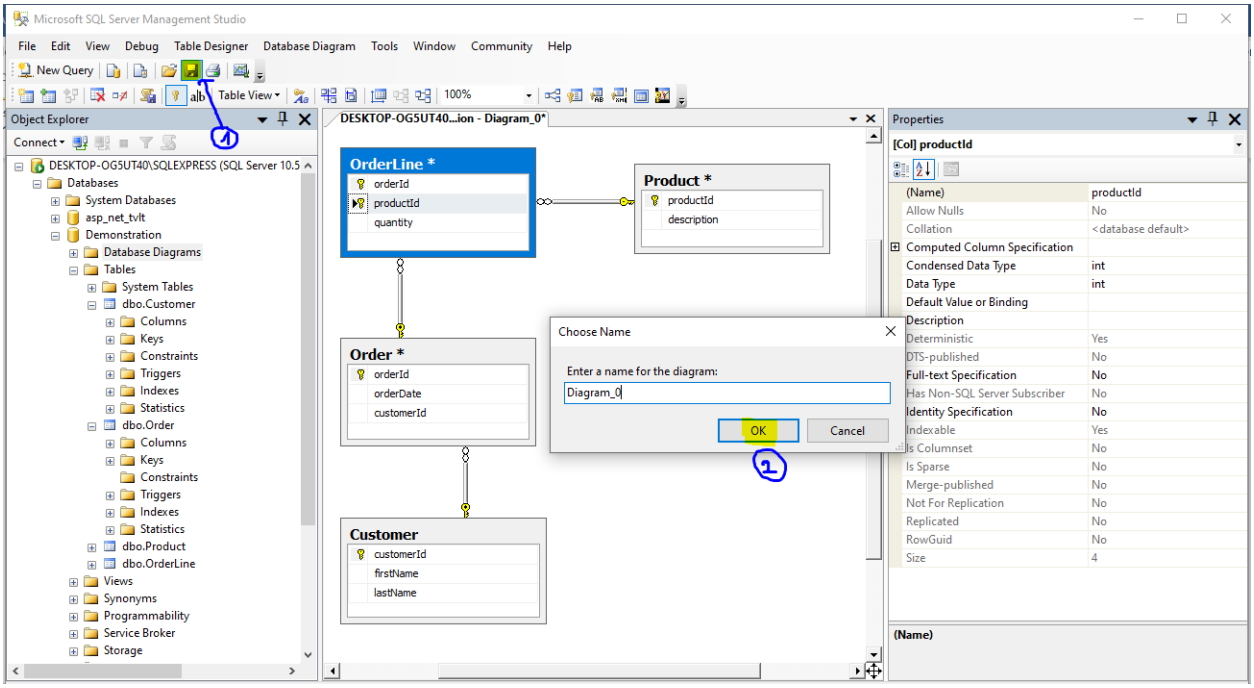
(A composite of orderId and productId is a primary key)



(Each individual productId or orderId is a foreign key)







(Save the Database Diagram)

Try to imput some data into the tables

Customer		
customerId	firstName	lastName
1	Emily	Phan
2	Aaron	Bergman
3	Aaron	Hawkins

Order		
orderId	orderDate	customerId
1	7/6/2016	1
2	3/14/2017	2
3	6/21/2017	3

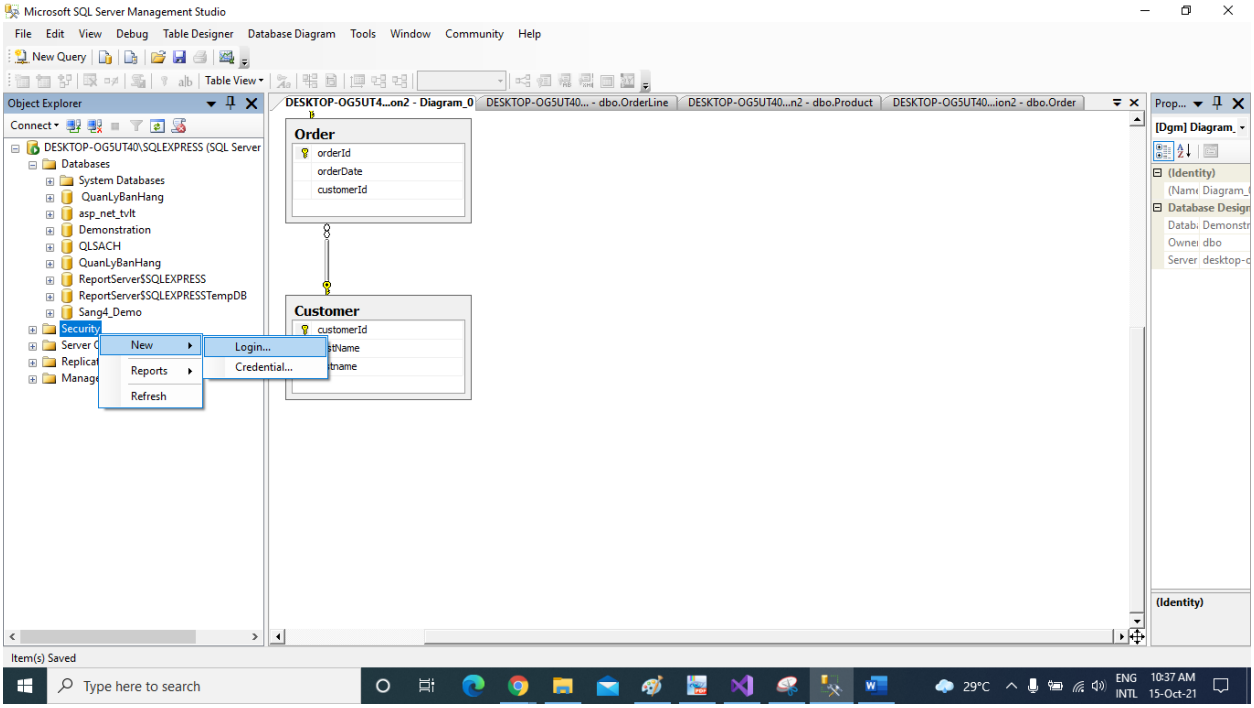
Product	
productId	description
1	Imation Secure
2	Xerox 2
3	Staple Envelope
4	Xerox 1964

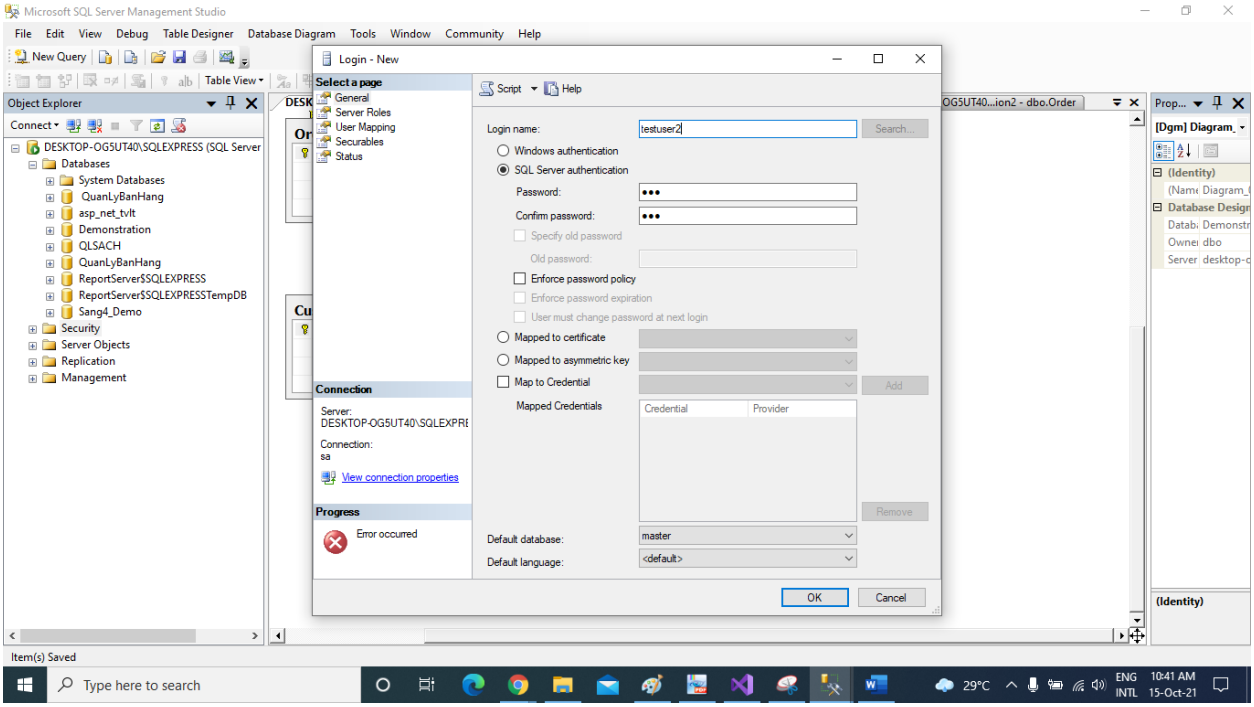
OrderLine		
orderId	productId	quantity
1	1	10
1	2	5
2	1	15
2	2	8
2	4	30
3	3	20

(Some sample data http://insightsthroughdata.com/customer-cohorts-and-product-sequence-with-orders-data/?utm_source=rss&utm_medium=rss&utm_campaign=customer-cohorts-and-product-sequence-with-orders-data)

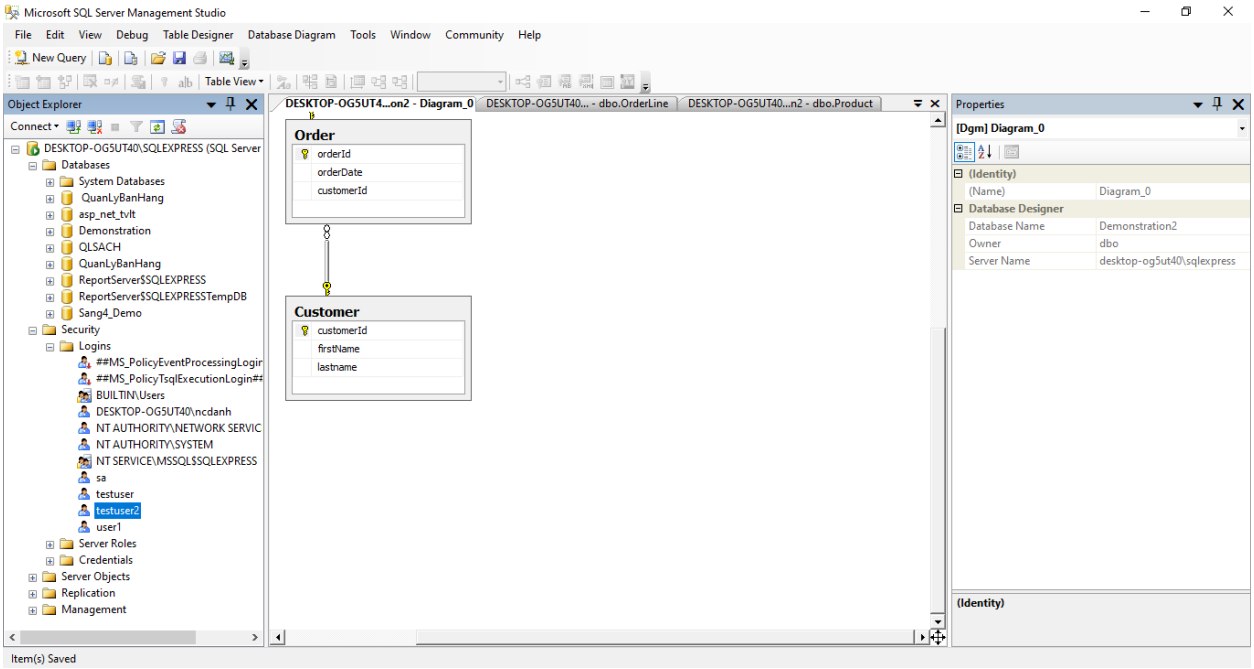
3. Create a new user and role

Create a new user

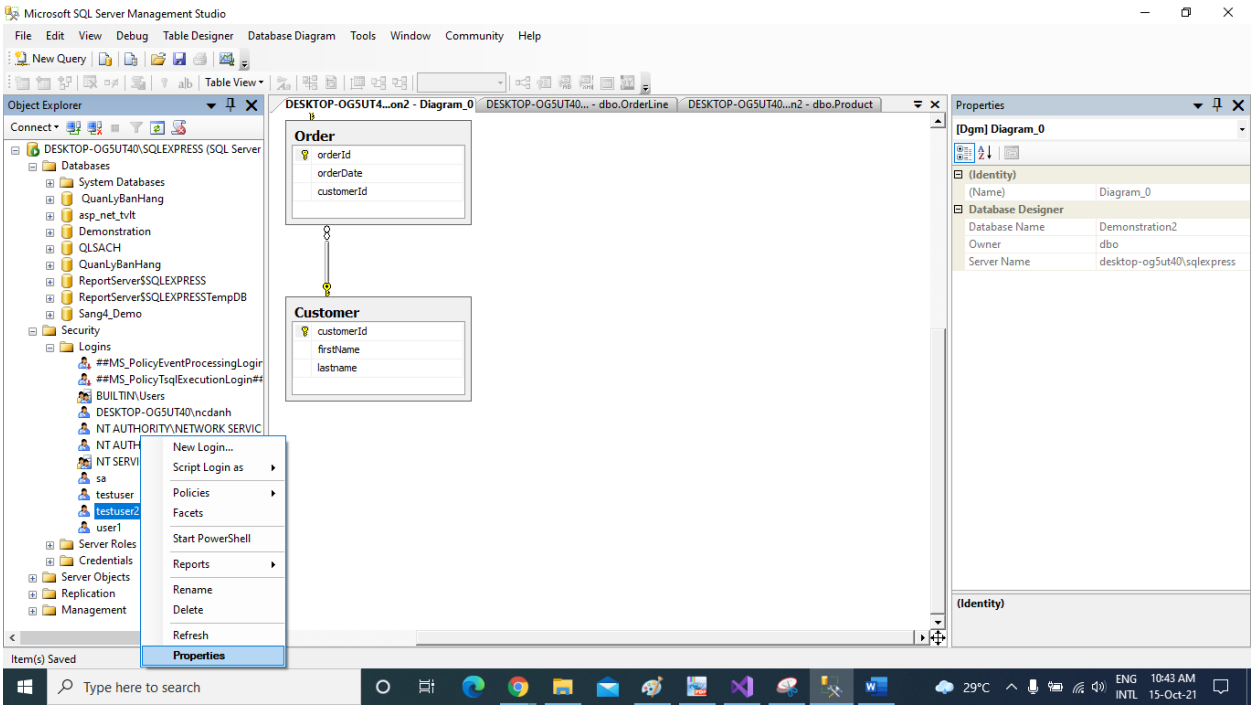


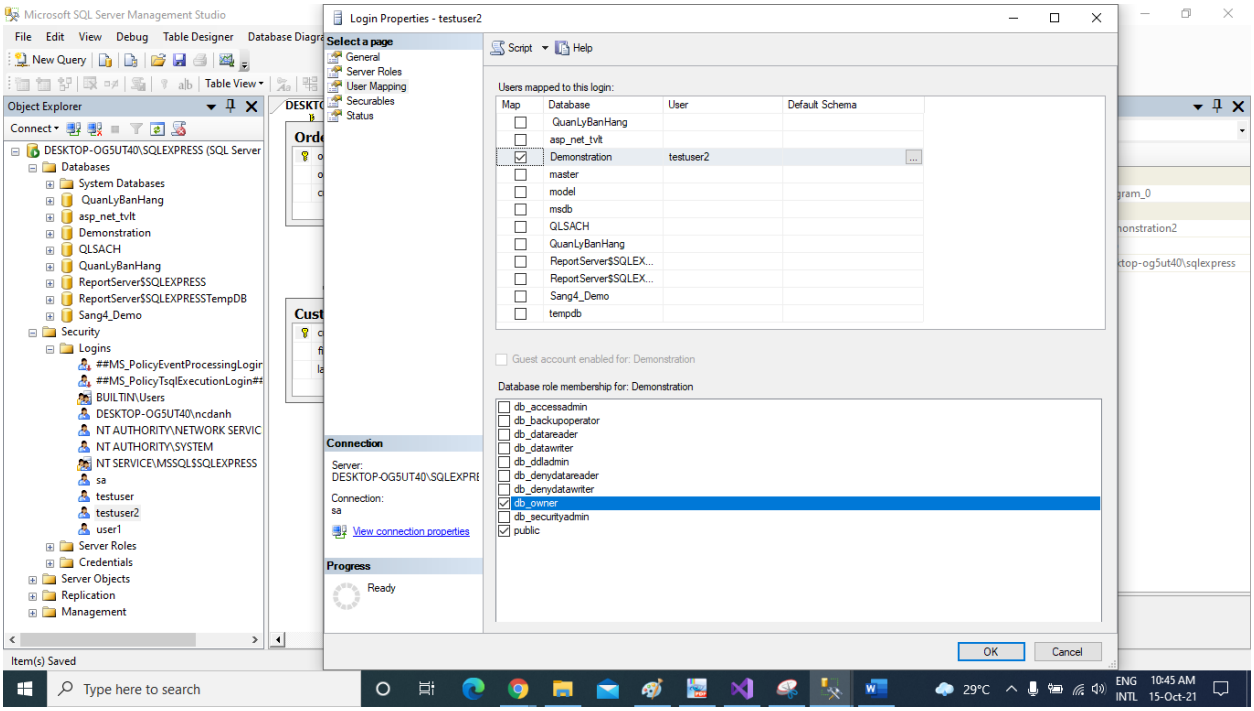


(Login name: testuser2; SQL Server authentication => password: 123; don't check Enfored password policy => OK)

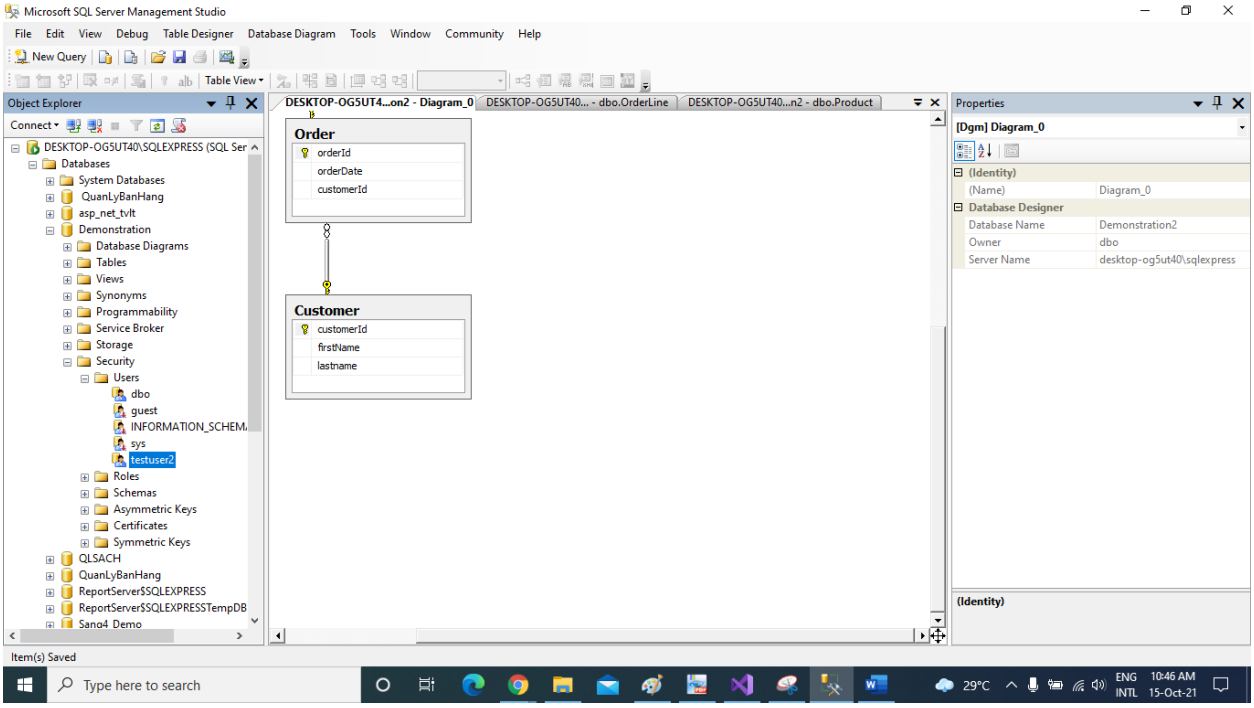


Assign a role for the user





(User mapping => Users mapped to this login: Demonstration; Data role membership for: Demonstration: db_owner =>OK)



We can connect to the Demonstration database via the testuser2: “User id = testuser2; password=123”.

/*

Error: Can't change table design in SQL Server 2008

https://stackoverflow.com/questions/9870968/cant-change-table-design-in-sql-server-2008/9870989

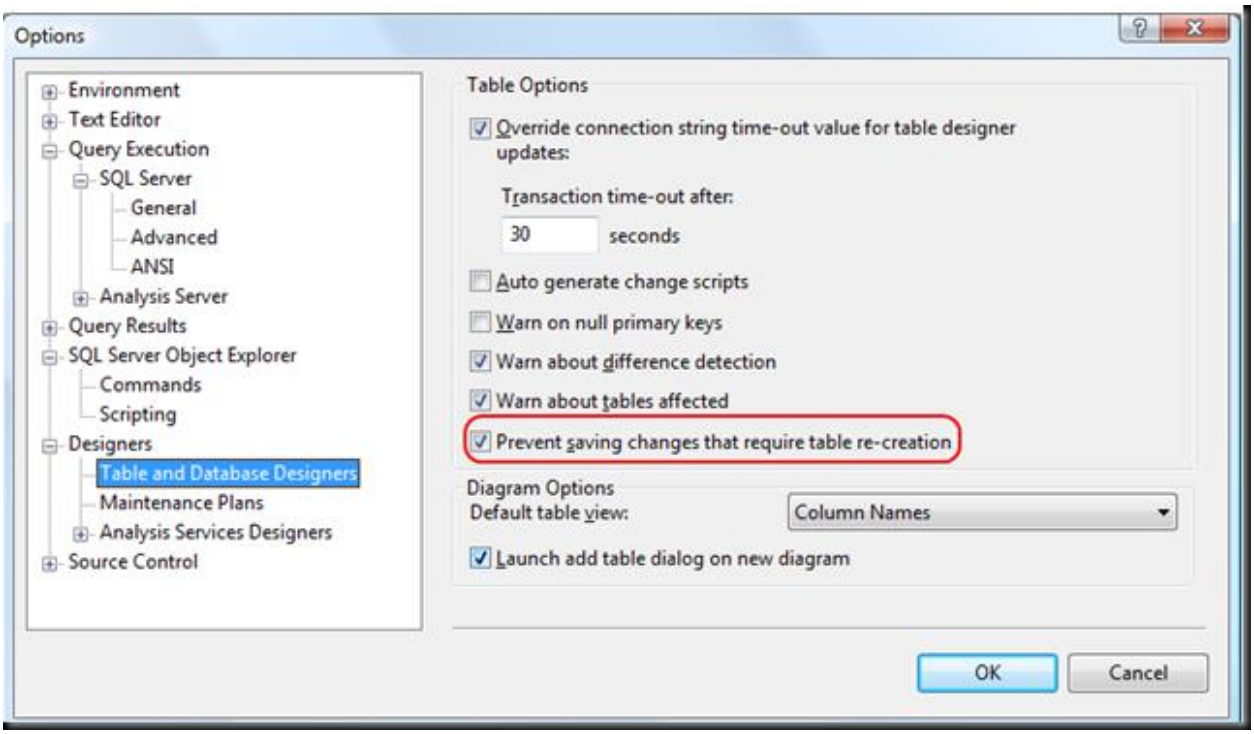
To change the Prevent saving changes that require the table re-creation option, follow these steps:

1. Open SQL Server Management Studio (SSMS).
2. On the Tools menu, click Options.
3. In the navigation pane of the Options window, click Designers.
4. Select or clear the Prevent saving changes that require the table re-creation check box, and then click OK.

Note If you disable this option, you are not warned when you save the table that the changes that you made have changed the metadata structure of the table. In this case, data loss may occur when you save the table.

Risk of turning off the "Prevent saving changes that require table re-creation" option

Although turning off this option can help you avoid re-creating a table, it can also lead to changes being lost. For example, suppose that you enable the Change Tracking feature in SQL Server 2008 to track changes to the table. When you perform an operation that causes the table to be re-created, you receive the error message that is mentioned in the "Symptoms" section. However, if you turn off this option, the existing change tracking information is deleted when the table is re-created. Therefore, we recommend that you do not work around this problem by turning off the option.



Error: Can't create a table with the same name I deleted before

https://dba.stackexchange.com/questions/135186/cant-create-a-table-with-the-same-name-i-deleted-before

Disconnect SQL Server Management Studio from the database and re-connect helped in my case.^

Note, there is also a "Intelli"-Sense cache which seems to get out of sync easily. To refresh that cache:

Press CTRL+SHIFT+R Go to Edit >> IntelliSense >> Refresh Local Cache

Another way:

After deleting the table, try closing SQL Mgmt Studio and open it again. Then try to create the new table with same name.

*/

Create a Relationship in SQL Server 2017

https://www.quackit.com/sql_server/sql_server_2017/tutorial/create_a_relationship_in_sql_server_2017.cfm

How do I change the date format in SQL Server Management Studio?
Asked By: Vada Altimiras | Last Updated: 5th February, 2020

Category: **technology and computing** **databases**

4.7/5 (4,906 Views . 26 Votes)

You can **change** the default **date format** per user by selecting the default language for that user in **SQL Management Studio** > Security > Logins > {user properties} > Default language.

connection string with username and password in sql server

https://www.connectionstrings.com/sql-server/

```
Server=myServerAddress;Database=myDataBase;User Id=myUsername;Password=myPassword;
```